Battle Of The Blood Mimics

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Abstract

Photoacoustic (PA) imaging combines the optical absorption contrast of laser excitation with the acoustic resolution of ultrasound, enabling non-invasive visualization of blood flow and biomarkers. Standardised blood-mimicking fluids (BMFs) are needed to ensure reproducible testing and calibration of PA technology. This study compares two candidate BMFs, a 1% carbon-microsphere fluid mixture (CF) and a 3.24% polystyrene-microsphere fluid mixture (PF), to evaluate their performance in PA vector-flow imaging. Experiments were conducted across wavelengths from 680 to 880nm using a 2.95mm diameter vessel wall-less phantom. The PF achieved a better signal-to-clutter ratio (SCR) (−6.7 ± 1.3dB), whereas the CF delivered more accurate flow-velocity measurements (2.07 ± 0.30mm/s, within 7% of expected). These findings reveal a trade-off between signal quality and quantitative flow accuracy. While the CF offers better flow measurements, the PF provides superior imaging clarity. This study demonstrates why the development of standardised and reliable BMFs for PA systems is critical to future clinical translation and highlights the key trade-offs between potential BMF candidates.

1. Introduction

Photoacoustic imaging (PAI) is an emerging imaging technology which pairs the clarity of optical-absorption techniques with the penetration depth of ultrasound (US) [1]. PAI uses short near infrared light pulses (600-900nm) directed into tissue to produce acoustic waves from tissues of interest such as haemoglobin. These light pulses are absorbed by haemoglobin in red blood cells (RBCs) 2-3 orders of magnitude more than the surrounding tissue [5]. The absorption leads to a small but rapid thermoelastic expansion of the haemoglobin creating acoustic waves in the tissue [2,3,4,5]. Due to acoustic waves experiencing 2-3 orders of magnitude less scattering than light they enable PAI to happen at depths of up to 9cm [1,6]. These acoustic signals are received by an ultrasound transducer at multiple angles which through data processing can measure flow speeds of blood down to 0.04mm/s [2,7]. In addition, PA is spectroscopic in nature, by analysing their spectral signatures researchers can determine physiological parameters such as blood-oxygenation and glucose level [1,8]. The high contrast, deep penetration, flow speed sensitivity, and spectroscopic nature of PA demonstrates its potential to revolutionise medical imaging for research and clinical applications.

PA positions itself as an information dense, non-invasive imaging technology that can be easily used by doctors to help diagnose and treat patients. PA can be used to identify adverse blood flow abnormalities such as clots, issues within microvasculature, and cancer/tumor supporting angiogenesis [1,2,3,9]. It can also be used for monitoring patient improvement with issues such as burn healing, tumor treatment, diabetic ulcer healing, and stroke recovery [10,11,12]. With only 4 commercially available systems world wide it is clear that PA must be developed further before it is accepted into clinical practice [13].

To support the advancement of PAI, further testing and refinement of experimental materials is required. This study aims to determine which of two blood-mimicking-fluids (BMFs), a 1% carbon-microsphere particle fluid mixture (CF), or a 3.24% polystyrene-microsphere particle fluid mixture (PF), provides the most reliable performance in PA experiments. By comparing the signal-to-clutter ratio (SCR), flow speed accuracy, and ease of use under a range of lasing wavelengths, this report aims to inform the selection of optimal testing materials and conditions. This research will contribute to the development of a robust testing protocol for future PA testing and development.

1.1 The Blood Mimicking fluids

CFs exhibit strong optical absorption due to their black pigmentation and glassy surface, producing a higher PA signal response and offering a cost-effective testing medium. However, these particles tend to fall out of suspension rapidly, and their reflective, glassy surfaces can also introduce image noise. Their variable particle size of 2-12µm compared to RBCs (7.5 to 8.7µm) also limits physiological accuracy [14,15,16].

In contrast, PFs more closely mimic the optical scattering behavior of blood. The red pigment and non-glassy surface of the spheres reduce reflection-induced noise. This fluid also maintains uniform suspension during imaging. Despite providing more realistic optical behavior, polymer-based fluids typically yield weaker PA signals due to their non-glassy nature, can stain experimental phantoms, and are significantly more expensive than the carbon alternative [17].

2. Theory

To compare the performance of the BMFs, it is essential to understand the principles of Doppler shift, velocity flow calculations, and SCR. These parameters are used to evaluate how accurately each fluid replicates blood flow characteristics in photoacoustic imaging.

The Doppler shift describes the frequency change of an ultrasound signal reflected from moving particles within the medium. The maximum detectable velocity before aliasing occurs is given by:

$$V_{\max} = \frac{F_{nyq}\, C}{\cos(\alpha - \phi)\, F_0} \tag{1}$$

Where V_{max} is the maximum velocity through the blood vessel that won't be aliased due to the Nyquist theorem. F_{nyq} is the Nyquist frequency, half of the frequency of the laser. C is the speed of sound through the medium. ϕ is the receiver angle and α is the vein angle from the normal to the US transducer. Lastly, F_0 is the center frequency of the US transducer. This equation calculates the V_{max} for our pump speed calculations based on known values. It also is the expected average velocity of the BMFs at the related pump speed. The expected Doppler frequency shift for a given flow velocity can be expressed as:

$$F_d = \frac{V \times \cos(\alpha - \phi) \times F_0}{C} \tag{2}$$

Where F_d is the expected doppler shift for a given receiver angle. V is the velocity of the haemoglobin through the blood vessel. This equation provides a theoretical baseline against which measured shifts from the BMFs can be compared to assess accuracy. The size of this shift also depends on the velocity of the source so this equation can be rearranged to determine the flow speed of a fluid based on its frequency shift.

$$\text{Pump rate} = V_{\max} \times \frac{\pi d^2}{4} \times [\text{conversion constants}] \tag{3}$$

This is the equation for the pump speed used in testing. Where ‘d’ is the diameter of the vessel in the phantom and conversion constants are used to ensure the outcome is ml/minute.

Finally, the SCR quantifies the proportion of the doppler signal from moving particles to the background “clutter” produced by stationary particles. Because clutter peaks are much larger than the doppler signal, SCR is expressed in decibels (dB). A less negative SCR value indicates a stronger doppler signal relative to the clutter present and therefore improved detection performance.

2.1 Predictions and Hypothesis

Table 1: *Known parameters*

Parameter	Value
Pulse Repetition Frequency (PRF)	20 Hz
Nyquist Frequency (F_{nyq})	10 Hz
Centroid Frequency of the Transducer(f_0)	7.6 MHz

Receiving Angles (ϕ)	-10°, -5°, 0°, 5°, 10°
Speed of Sound through phantom (C)	1449.3 ± 3.37 m/s [18]
Vein angle from the normal to the ultrasound transducer (α)	90°

To determine which BMF is best the following predictions were made to compare to the results:

Predictions were made for the max velocity using all receive angles, below is an example for a 10° receive angle. V_{max} is smallest at the largest receive angle, thus ±10° results in our limit for aliasing to occur.

$$V_{\max} = \frac{F_{nyq}\, C}{\cos(\alpha - \phi)\, F_0} = \frac{10 \text{ Hz} \times 1449.3 \text{ m s}^{-1}}{\cos(90° \pm 10°) \times (7.6 \times 10^6) \text{ Hz}} = 10.98 \text{ mm s}^{-1}$$

The pump speed should be set to deliver at most half the max velocity of flow to ensure no aliasing occurs in the results due to the Nyquist theorem. In addition, this study's primary focus is slow blood flow as it is the primary focus of many PA researchers. The pump speed was set to deliver a velocity closer to that experienced in a small vessel, putting the expected average velocity far from aliasing. 1.936 mm/s was chosen as a value that represented the slow blood flow in tissue that would likely be of interest in future clinical studies [19,20].

$$\text{Pump rate} = 1.936 \text{ mm s}^{-1} \times \frac{\pi \times 2.95^2 \text{ mm}}{4} \times \left[60 \frac{\text{s}}{\text{min}} \times (1 \times 10^{-3}) \frac{\text{cm}^3}{\text{mm}^3}\right]$$

$$\text{Pump rate} = 794.1 \pm 3.2\ \mu\text{L/min}$$

The expected doppler shift was calculated for each receive angle, below is an example at ± 10° receive angle.

$$F_d = \frac{V \times \cos(\alpha - \phi) \times F_0}{C} = \frac{1.936 \text{ mm s}^{-1} \times \cos(90° \pm 10°) \times (7.6 \times 10^6) \text{ Hz}}{1449.3 \text{ m s}^{-1}} = \pm 1.763 \text{ Hz}$$

Lastly, this study predicts that the CF will have a larger SCR due to its optical properties producing a larger signal.

3. Literature Review

The earliest documentation of the PA effect being used in technology was 1880 by Alexander Bell. After the invention of the telephone Bell decided to turn his attention to light transmitted information [21]. By June 1880 Bell and his assistant Charles Sumner Tainter were able to transmit voice messages roughly 213 meters using focused sun rays and a vibrating mirror as their 'laser'. This was done using the photosensitivity of selenium as a receiver which was connected to a powered telephone circuit. Through the research on his "Photophone" Bell made the key discovery that; by illuminating solids with a pulsed beam of light, the energy created an acoustic wave at the same frequency as the pulses [21]. This discovery astonished the scientific community of the time but as communication through radio developed through the early 20th century the idea of a Photophone was all but forgotten.

Revived in the late 1930s for infrared gas analysis, the photophonic effect saw brief renewed interest before being replaced by more sensitive techniques like gas chromatography and infrared spectrophotometry in the late 1940s [21]. Once again, after reduced interest for decades, the photophonic effect had a resurgence in the 1960-70s due to the discovery of laser technology in the 1960s. Lasers provided a power and stable light source which paved the way for PA spectroscopy in the coming decades [22]. The earliest case of biomedical applications for photoacoustics was in 1964 by French scientist Amar et .al [21]. They discovered ultrasound signals were generated in the occipital bone of a rabbit when low power light (non-injuring) was pulsed through the rabbits eye. At the time Bell labs focused on the interactions of light with solids. This is when Rosencwaig coined the term 'Photoacoustics' to be used referring to the phenomenon interacting with solids in his 1973 paper [23]. Early biological discoveries using PA followed, including, confirming early hypotheses about the proteins used in cataract development [21].

As PAI advanced two modalities emerged: photoacoustic computed tomography (PACT) and photoacoustic microscopy (PAM). PACT and PAM provide a trade-off between depth and resolution. PACT uses diffuse illumination and parallel detection with a transducer array, which is ideal for deep, real-time preclinical imaging, such as tumors, despite its low resolution (200–500 μm). In contrast, PAM offers high-resolution, microscopic detail. It uses a confocally aligned optical-acoustic setup that mechanically scans point-by-point, deriving lateral resolution from the joint foci and depth information from acoustic time-of-flight [22]. Both of these modalities are used for tracking blood flow, this study uses PAM as it focuses on blood flow and BMFs which need microscopic detail. The first records of using the doppler shift technique to track blood flow that this study could find was by Lihong V. Wang and his team in 2008 [24]. In comparison blood flow has been tracked by ultrasound since the early 1960s, this shows how young PAI of blood flow is [25]. Throughout history researchers have used non-stardised BMFs, reducing the reproducibility, clarity and capability for collaborative work thus slowing PA technology development.

J. J. Riksen et al. explained in 2023 that "for PA to transition from a 'nice-to-have' to a 'need-to-have' modality" it will require clinical trials focusing on the value for patients and clinicians [4].

This is why clinical studies are currently focusing on the use of PA to address gaps in identification of cancer [26,27,28,29,30]. For PA to solidify itself as an imaging modality it must address this niche better than existing options, likely through: cost effectiveness, portability, increasing the success rate of identification, and ease of operation in many healthcare settings [31]. Current leaders in PAI, such as Luxonus from Japan, are focused on using the increased resolution and ease for patients to create a commercially viable product able to be installed in offices for use in advanced diagnosis and monitoring [32]. The LME-01 from Luxonus uses a unique semi sphere of ultrasound transducers allowing for signals to be received at greater angles, enabling vessels traveling horizontally or up to 60° from horizontal to be imaged, creating highly detailed images [33]. Seno medical's Imagio is a PA breast cancer diagnostic tool that has been FDA approved [34]. This approval allows PA technology to be in the hands of doctors and help expose patients to PA technology, this will help the transition in clinical settings for future PA companies. The Imagio imaging device focuses on being an effective clinical tool by using a familiar shaped ultrasound wand and simply overlaying a vessel map over an ultrasound image to assist in the identification of tumors in patients' breasts. Vevo F2 LAZR-X20 is a preclinical research tool focusing on versatility, by tuning the laser it can have a max resolution of 30 micron or an image depth of up to 9cm [6]. All of these leaders are focused on blood flow, and blood oxygen vessel mapping to focus on cancerous tumor identification and monitoring.

Development is ongoing for LED powered PAI technology which delivers a safe and portable monitoring device, recording things such as; blood biomarkers, arthritis, and cancerous growths [35,36]. This is promising for PA as it means in some instances LEDs can be used which are much cheaper, and reserve the expensive high powered lasers for more specialised diagnosing situations. It has been shown that CNM-based nanocomposites can be used as injectable contrast agents to broaden the utility of PAI by absorbing light across the UV, visible, and NIR-I/II windows. This in turn enhances image resolution, increases penetration depth, and enables more versatile multispectral analysis allowing for the validation of spectroscopic techniques [37]. This emphasized photoacoustic intensity has proven effective for in vivo monitoring applications like drug delivery and tumor cell targeting [38]. This shows that for cases which need more intensity, PA can adapt to be a semi-invasive modality to fulfil those needs. Lastly, future advancements with machine learning and AI appear promising for increasing diagnostic accuracy but need further development [39].

BMFs are very important as they offer a standardised and repeatable test that can be used to calibrate and compare machines. Calibration is required for any tool entering clinical use as inaccurate/uncalibrated tools can threaten the safety of patients as they can lead to misdiagnosis [40]. Table 2 is used when making ultrasound mimics, so it is missing the crucial component for PA - optical absorption - And research has shown its values do not exactly match with those of human blood [41].

Table 2: *Properties and the values that BMFs should aim for as defined by the International Electrotechnical Commission[42].*

Properties of BMFs	Values
Viscosity ($\times 10^{-3}$ Pa·s)	4.0 ± 0.4
Attenuation (dB/cm/MHz)	≤ 0.1
Acoustic speed (m/s)	1570 ± 30
Density ($\times 10^{3}$ kg/m^{3})	1.050 ± 0.040

There have been a variety of BMFs used in history; in 1994 india ink was used in an early phantom to demonstrate the possibilities of PA in medical imaging [21]. Since then limitations have been found with india ink, such as brighter signals at the edges of vessels, and providing flat absorption spectra [43,44]. Nano particles in suspension are now used widely in PA research to emulate RBCs in blood. There is limited research on direct comparisons between BMFs and human blood. This leaves it unclear if the BMFs are improving in terms of accuracy to theoretical predictions or accuracy to how human blood would behave. Both carbon and polystyrene micron sized spheres have been used in suspension to mimic blood [45,46]. These fluids must attempt to match the properties mentioned in table 2 in addition to having large SCR, accurate flow speed measurements and stability in suspension. Another unconsidered aspect when using micron spheres is the shape, RBCs are a biconcave disc (**Fig. 1.0)** (7.5 to 8.7µm diameter and 1.7 to 2.2µm in width) which means light would be interacting with it differently based on its orientation, unlike a sphere [15][42]. Even though there are three main cells in human blood, platelets (~2µm), white blood cells (WBC) (12-15µm) and RBC [47,48,49]. PA light waves are tuned to be absorbed mostly by haemoglobin which is only present in RBC, thus the artificial particles should focus on replicating RBCs and not other cells [50]. Future advances could include manufacturing red polystyrene biconcave discs to better mimic the optical and acoustic properties of RBCs.

Despite progress in photoacoustic imaging technology, several critical gaps remain in blood mimicking fluid development that limit reproducibility and standardization of PAI research.

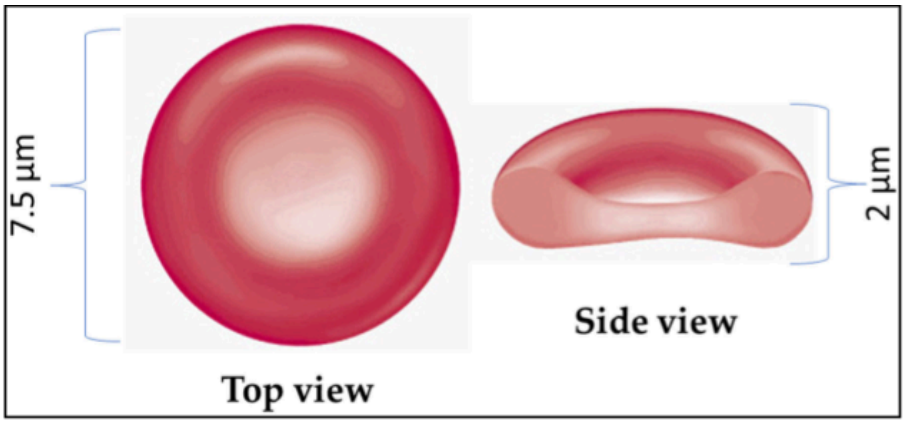

Figure 1.0: *Shape and dimensions of a red blood cell present in human blood fluid.*

A major challenge is BMF stability during experiments. When the density of particles is more than that of the surrounding liquid, settling may happen, which can be visible at low flow speeds [16]. Polystyrene scatter particles can precipitate to the bottom when their density is greater than the mixture fluid [42]. This density driven settling creates concentration gradients that require constant remixing and introduce measurement inconsistencies.

There is currently no universal standard for BMF composition across laboratories. Current standardised methods for system testing are lacking, impeding objective device performance evaluation, calibration and inter-device comparisons [51]. Different groups use vastly different formulations, from india ink solutions to polystyrene microspheres to custom nanoparticle suspensions, making cross-study comparisons difficult. IPASC has developed recommendations for ideal phantom characteristics to help standardize practices and improve system calibration across the field [51]. Without agreed-upon reference fluids with well characterized properties, it's difficult to establish benchmark performance metrics or determine whether reported differences in system performance reflect actual technological advances or simply variations in test fluids.

Real blood shows wavelength-dependent absorption that changes with oxygenation state, which is fundamental to many PA applications. To test this, it has been found that using different concentrations of intralipid (a fat emulsion) can mimic both deoxygenated and oxygenated blood [51]. Several general limitations have been identified in prior phantom work - tissue phantom optical and acoustic properties were often not characterized or were not biologically relevant, and test methods using ex-vivo tissues as phantoms lack reproducibility [53]. Linear spectroscopic inversions are widely used in photoacoustic imaging to estimate blood oxygen saturation, but they don't account for the spatially varying wavelength-dependence of the light fluence within tissue, introducing spectral coloring as a potentially significant source of error [54].

Using animal blood to mimic human blood comes with a number of issues, including the approval from ethics committees creating a barrier for use [55]. There is extensive research about safety precautions when using animal blood as mishandling it can result in disease, hospitalization, or death [56]. Both animal and human blood possess a shelf life limited by the degradation of the haemoglobin proteins they contain and

the viscosity which affects flow speed. The shelf life varies drastically depending on storage conditions and what animal blood is taken [57,58]. Future research into artificial blood products could pose another option for tests as they have extended shelf life, up to 2 years at room temperature [59]. Lastly, animal blood has different sized RBC and haemoglobin concentrations to humans which is pivotal for the signal strength in PAI [60,61].

There is no agreed upon standard for BMF testing. Comparing flow speeds and doppler shifts to expected values is an effective analytical way of comparing fluids, but comparisons to human blood results would be ideal. It has been shown that using the same poly-particle fluid as this study with a range of vessel widths the flow results consistently underestimate the known velocities [45]. Tests for blood flow using BMFs use a variety of experimental setups, from tubes suspended in water to more ideal wall-less phantoms [43,45]. Each method can lead to varying results highlighting the importance of standardisation of both phantoms and fluids.

BMFs are essential for PA development as they allow for consistent, repeatable, and collaborative experiments. The lack of research into BMFs is evident in relation to PA technology. Without standardisation of accurate and stable BMFs and phantoms researchers are unable to relate their data to those around them, limiting the pace of progression towards clinically viable PA technology. Future progress will depend on the creation of reference BMFs that replicate the optical absorption and acoustic properties of real blood, enabling reliable calibration and clinical translation of PA systems.

4. Experimental Methods

4.1 Phantom Fabrication

The initial step involved fabricating phantoms for vector flow testing. Sections of tin-foil were cut and shaped into rectangular moulds by folding up the corners and securing them with staples. Small holes were punctured on opposite sides of each rectangle using a copper wire at a depth of ~18mm, the focal distance of the US transducer. Connectors were inserted into these holes using small amounts of force and slowly rotating them. A copper wire was then inserted into the moulds by passing it through the connectors at each end. Three moulds were prepared: two containing 1 ± 0.005mm diameter wires and one containing a 2.95 ± 0.005mm diameter wire. The latter was constructed with one connector due to the wire being too large to pass through it, and protruded without a connector at the opposite end as shown in Figure 2.0.

An oven was preheated to 200°C. During the heating period, the phantom material (Humimic SimuGel™ #0) was cut into small cubes and placed in glass beakers for melting [62]. Once the oven reached 132°C , the beakers containing the cubes were placed inside and left for approximately 30 minutes or until fully melted. The molten gel was then poured into the prepared molds. These were immediately returned to

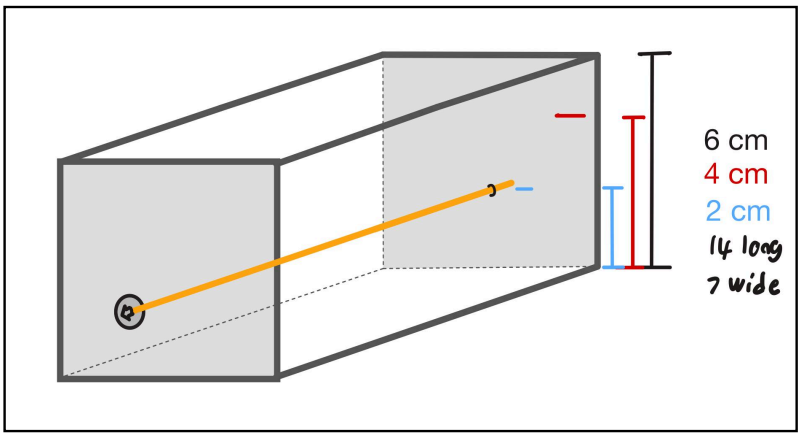

Figure 2.0: *Schematic of the phantoms used to mimic human tissue, with a hollow "vessel" running through the middle of the phantom mimicking a typical blood vessel.*

the oven for one hour or until all bubbles had escaped the gel or were at the surface. If bubbles were present at the surface they were popped with a fine needle.

Once the bubbles were removed, the moulds were placed on the bench to cool overnight. Once cooled, the tin-foil was carefully peeled away from the phantoms. The metal wire was then slowly extracted from the center of each phantom using pliers to grip the rod. The resulting product was a rectangular phantom with connectors and a hollow "vessel" running through the middle, as shown in Fig. 2.0. This channel would serve as the conduit for fluid flow during testing.

4.2 Experimental Setup

All photoacoustic vector flow measurements were conducted using the 2.95 ± 0.005mm vessel phantom described in Section 3.1. The experimental configuration is illustrated in Figure 2.1.

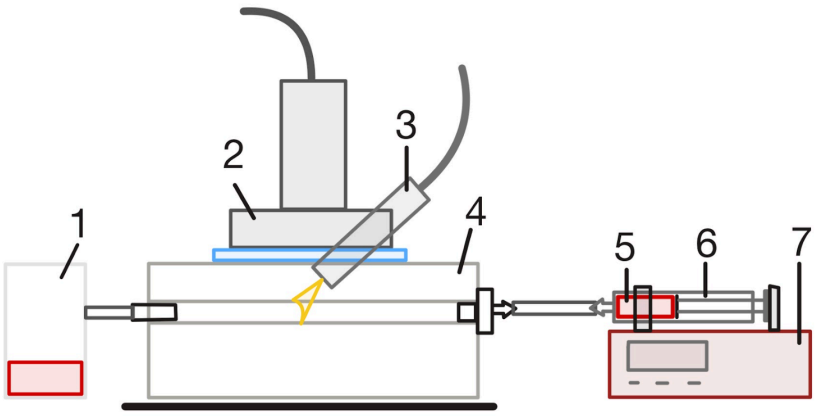

Fig.2.1: *Diagram of Experimental Setup.*

4.2.1 System Components

The experimental apparatus consisted of seven primary components (Fig. 2.1):

1. Fluid reservoir - A beaker positioned to collect outgoing fluid from the phantom.

2. Ultrasound transducer - Used for ultrasound signal detection and imaging.

3. Laser system - Opotek RADIANT 532 LD Tunable Optical Parametric Oscillator Laser, provided lasing at wavelengths ranging from 680nm to 880nm.

4. Flow Phantom - As described in section 3.1, containing the channel mimicking a blood vessel.

5. Blood-mimicking fluid - One of two blood mimicking fluids used, 1% carbon sphere particle fluid or 3.24% poly sphere particle fluid.

6. Fluid delivery syringe - Connected to the phantom inlet.

7. Syringe pump - Programmable pump for precise flow rate control.

4.2.2 Flow Rate Calibration

The syringe pump was set to deliver fluid at a rate of $794.1 \pm 3.2 \mu L/min$. This flow rate was calculated using the equations stated in the theory section. The internal diameter of the syringe vessel (12.25 ± 0.005 mm) was required to calibrate the pump in its menus.

4.2.3 Laser Alignment Procedure

Prior to alignment of the laser the power produced by the laser at each wavelength was measured.

The laser was aligned to maximise the PA signal from the vessel within the phantom. With the fluid stationary in the vessel and the laser lasing at 680nm, the laser position was adjusted repeatedly while monitoring the peak signal amplitude. This was done until a maximum intensity was identified, indicating optimal overlap between the laser beam and the vessel. Once in position, the laser was secured to maintain alignment throughout the experiment.

The optimal alignment yielded peak signal amplitudes of approximately 220 bits for the poly-sphere fluid and 8000 bits for the carbon-sphere fluid, reflecting the difference in optical absorption between the two fluids.

4.2.4 Data Acquisition

For each blood-mimicking fluid, the following system was used:

1. The syringe was filled with the test fluid and placed in the syringe pump.
2. The pump was activated to establish steady flow through the phantom.
3. Once stable flow was confirmed, photoacoustic data was acquired at six wavelengths, 680nm, 720nm, 760nm, 800nm, 840nm and 880nm.
4. At each wavelength, 1000 frames were recorded using the provided acquisition script.
5. After completing all recordings, and before the next fluid was used, distilled water was pumped through the vessel to flush the phantom of any residual BMF.

All components remained fixed in position throughout the experiment to ensure consistent measurements and geometry across all trials.

5. Data Processing

This section will outline the procedure undertaken to acquire all key results found within the experimental data. Throughout this experiment MATLAB was used for processing all data obtained for all wavelengths measured for each fluid to find all key values from this experiment. Python code was also used to analyse frequency spectrum graphs to obtain values for SCR and doppler shift.

5.1 Image Reconstruction, Frequency Spectrum Graphs and Laser Intensity

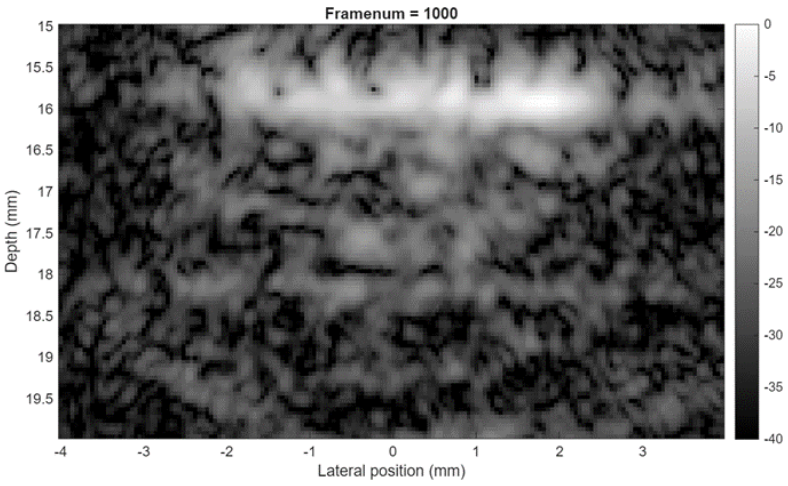

Figure 3.0: *Example image reconstruction of signals received via the transducer over all the receiving angles at each wavelength from 680-880nm.*

First each data set was compiled using the code file Reconstruct_JS_RunPA.m. Within this file the images were set to the transducer's parameters and the image grid was set to produce an image at depths 15 to 20mm and lateral positions of -4mm to 4mm relative to the transducer center. This produced a 1000 frame image reconstruction of the fluid flow within the phantom vessel as the chosen photoacoustic fluid was being pumped through:

The brighter areas represent regions of higher reflectivity from the laser signal. The depths of 15-15.75mm and 18.5-20mm represent the noise produced by the phantom structure and the region at depths of 16-18mm represent the inside of the vein where the photoacoustic fluid was present. The bright lines found between 15.5-16mm and 18-18.5mm depths, represent the vein walls. These are present due to the large change in impedance moving from the Humimic SimuGel™ #0 to the chosen photoacoustic fluid [62]. Lastly, this produced frames_param_(fluid name and wavelength).mat files that were used in all other code files to obtain results.

Next the maximum intensity present in our reconstructed image was found to determine laser intensity. By using Intensities_ Reconstruct_JS_RunPA.m, an altered version of the previous code to give the maximum intensity over the selected region. The region chosen was set at depths between 15.5-17.5mm and lateral positions set between -1.5-3mm, this can be seen as the bright region in Figure 3.0 which marks the top of the phantom vessel. The maximum intensity was found over this entire region and compared with the laser power found during the experimental procedure.

5.2 Doppler Shift Analysis

The frames_param_(fluid name and wavelength).mat files were then processed using the code file slow_time_spectrum.m to first isolate a desired region to obtain frequency spectrum graphs for each receiving angle. The chosen region was set as a 40 by 20-pixel box set with its top left corner set at pixel position (70, 36).

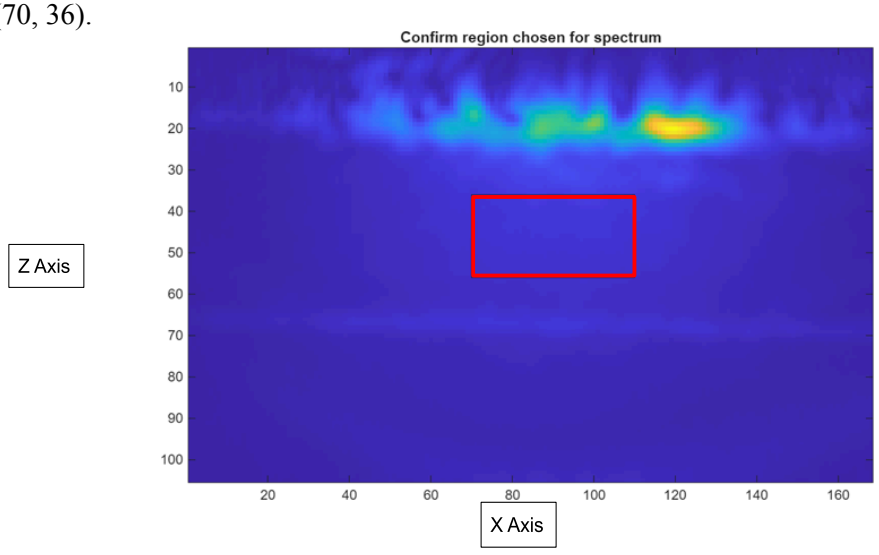

Figure 3.1: *Example image of chosen region, shown by the red box, to plot frequency spectrums of the amplitude at each receiving angle for each wavelength from 680-880nm.*

The brighter regions show, as also seen in Figure 3.0, the level of laser signal reflectivity due to impedance changes.

This code file then produced five frequency spectrum graphs each corresponding to a given receiving angle for this chosen region:

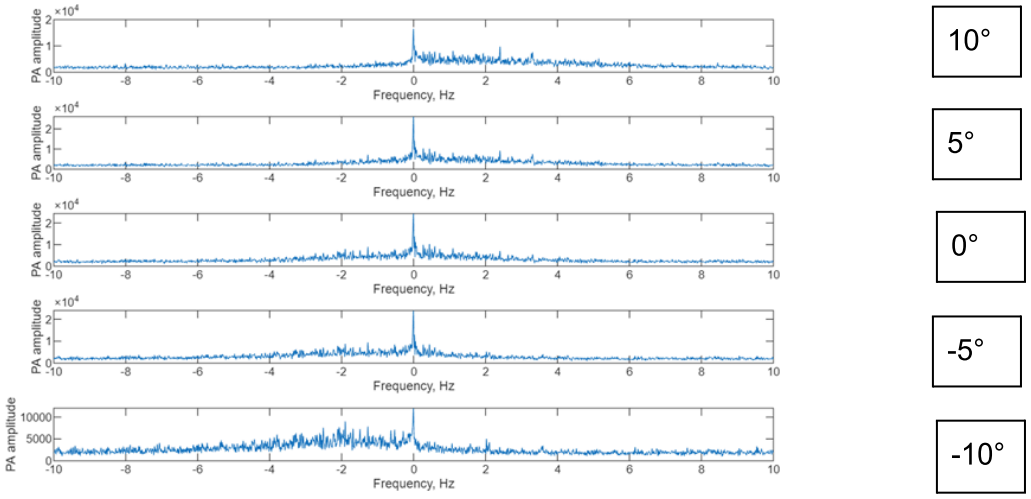

Figure 3.2: *Example frequency spectrum of the signal and clutter amplitude at each receiving angle from 10° at the top, to -10° at the bottom. The clutter peak is found around a frequency of 0Hz while the signal is found across the spectrum giving much lower amplitudes by comparison. Frequency ranges from -10Hz to 10Hz.*

The Peak at approximately 0Hz is the clutter peak which represents the noise produced by stationary particles within our chosen region. The surrounding signal envelope represents the signal produced by the photoacoustic fluid flowing through the phantom.

In order to analyse the signal envelope further the clutter peak must be filtered out. To do this a high-pass filter was used by integrating the code file highpass_GR_US.m to the slow_time_spectrum.m code file. The filter was set to act over the same region that the original spectrum was obtained from, with a cut-off frequency of 0.01Hz giving the following filtered frequency spectrums:

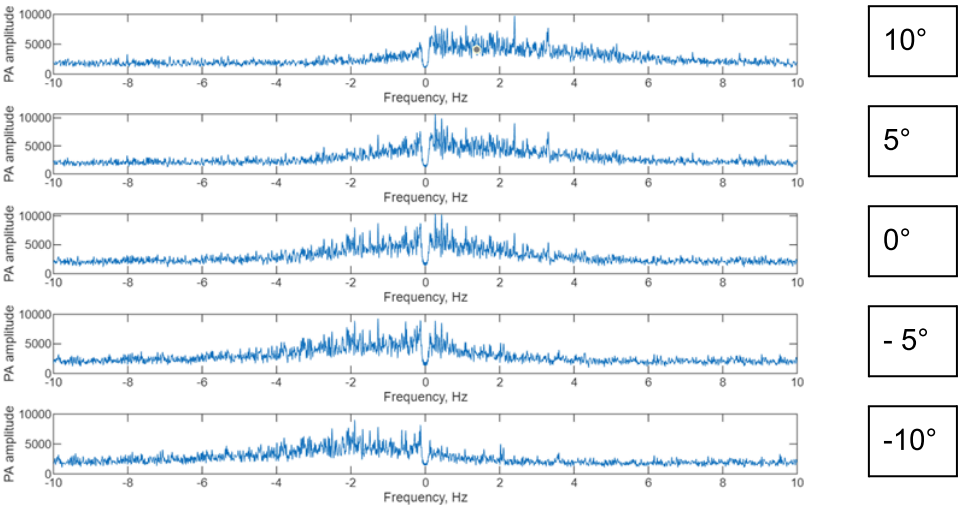

Figure 3.3: *Example 0.01Hz high-pass filtered frequency spectrum showing only the remaining signal envelope used to find signal amplitude and doppler shift. Frequency ranges from -10Hz to 10Hz.*

The data contained within each graph was then exported into Excel using the code file figure_to_excel_code.m, these values could then be used with python code to analyse for the Doppler shift frequency at each angle.

By reconstructing the filtered frequency spectrum using plt.ployfit, a precise estimate of the signal envelope of each data set was plotted. The peak of each curve indicated the frequency associated with the Doppler shift at that receiving angle:.

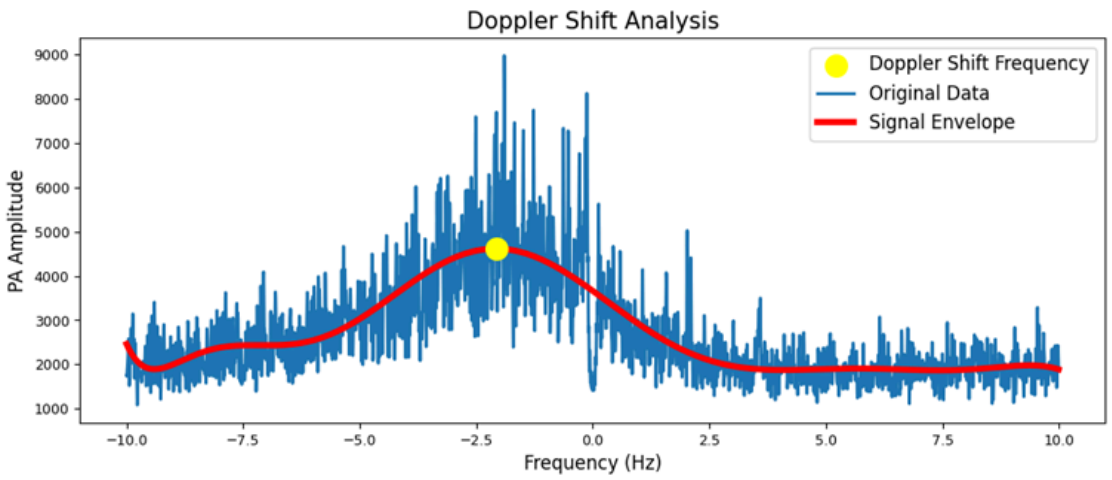

Figure 3.4: *Example reconstruction of 0.01Hz high-pass filtered frequency spectrum with envelope. The blue curve represents the original filtered data, the red line represents the fit for the signal envelope and the yellow dot represents the doppler shift frequency which is found by finding the peak signal amplitude.*

Doppler frequencies were recorded at every receiving angle for each data set. Next the average doppler frequency at each angle was found for both fluids. These frequencies were then compared to the expected doppler shift at the matching receive angle, which was found using equation 2.

5.3 Signal-to-Clutter Ratio (SCR) Calculations

By exporting the unfiltered doppler shift graph data to excel, shown in Figure 3.3, the data was reconstructed in Google Colab to find both the clutter peak amplitude and the signal amplitude for each data set:

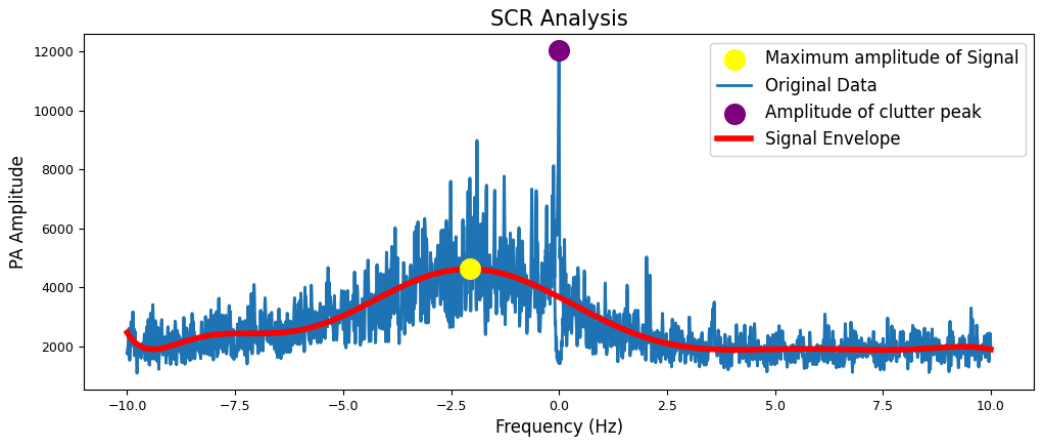

Figure 4.0: *Example reconstruction of unfiltered frequency spectrum with signal and clutter peaks marked by the yellow and purple dots respectively.*

To find the SCR of each frequency spectrum the amplitude values obtained for clutter peak and signal peak were used in the following formula:

$$(SCR)_{dB} = 10\,log\left(\frac{A_S}{A_C}\right) \tag{4}$$

Where $(SCR)_{dB}$ is the signal-to-clutter ratio scaled in decibels, A_S is the peak signal amplitude and A_C is the amplitude of the clutter peak. Each SCR value was averaged over all angles for each laser wavelength measurement, recorded for both photoacoustic fluids, including the standard deviation.

The SCRs found at each wavelength were then compared to see which fluid gave a less negative value meaning the signal envelope amplitude was closer to the clutter peak amplitude therefore an image that contained less noise.

5.4 Vector Flow Analysis

By processing the frames_param_(fluid name and wavelength).mat file using the code file Compute_Vectorflow.the mean and max velocities were found for each data set along with an associated standard deviation. First the code analysed all the given data in the frames_param_(fluid name and wavelength).mat file to obtain vector flow map for the reconstructed image:

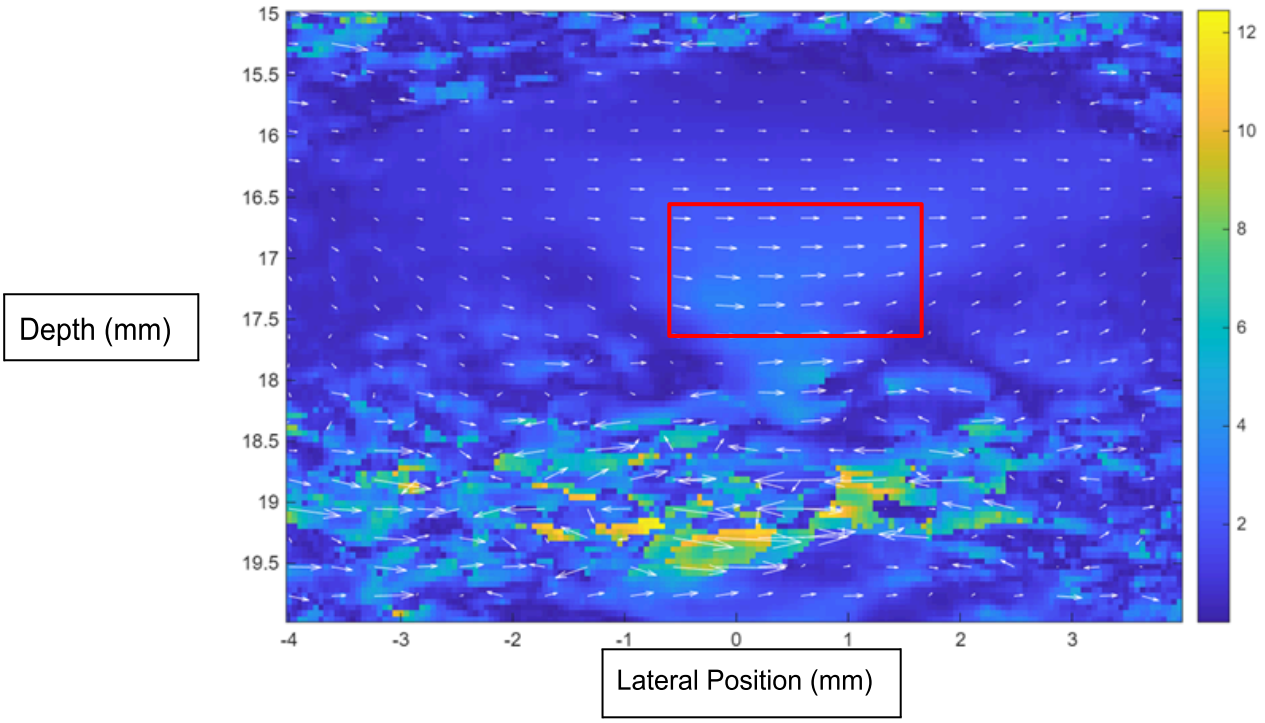

Figure 5.0: *Example image of a vector flow map detailing the magnitude and direction of flow in the phantom vein, with the chosen analysed region represented by the red box.*

With the arrows pointing in the direction of flow and the length of each representing the magnitude of the flow velocity. The region chosen to find the mean and max velocities of flow and give an associated standard deviation is shown by the 20 by 40 pixel red box at coordinates (70,36). These mean velocities were then compared at each wavelength for both fluids and to an expected mean flow velocity value. The maximum flow velocities for each fluid at each wavelength were then compared separately.

6. Results

These are the results obtained for all our key recordings during the experiment and data processing.

6.1 Doppler Shift, Laser Power and Intensity

The power of the laser prior to the recordings and intensities found at each wavelength for both fluids is given below:

Table 3.0: *Maximum signal intensity and power of incident laser on the phantom at each wavelength as found from image reconstruction data. The intensity is normalised against laser power to account for fluctuations.*

Wavelength (nm)	Laser Power (mW/cm^2)	Fluid	Signal Intensity	Normalised Signal Intensity
680	167	1% Carbon	6112	36.60
		3.24% Polyspheres	142.1	0.8509
720	163	1% Carbon	4751	29.15
		3.24% Polyspheres	304.2	1.866
760	156	1% Carbon	2589	16.60
		3.24% Polyspheres	132.4	0.8487
800	141	1% Carbon	4509	31.98
		3.24% Polyspheres	124.8	0.8851
840	117	1% Carbon	6539	55.89
		3.24% Polyspheres	87.63	0.7490
880	108	1% Carbon	6552	60.67
		3.24% polyspheres	59.53	0.5512

Due to the varying power of the laser at each wavelength the intensity is normalised to see which is best at producing higher intensities relative to the laser power.

The maximum signal intensities found using the CF are on average 49.8 times higher than those found with PF. These results are very inconsistent with each other. The highest ratio of maximum signal intensity to laser power found for CF was 60.67 at 880nm, and for the PF this was 1.866 at 720nm.

The following graph shows the average doppler frequency at each receive angle averaged over all wavelengths for both fluids. This was also compared to expected shifts calculated using equation 2:

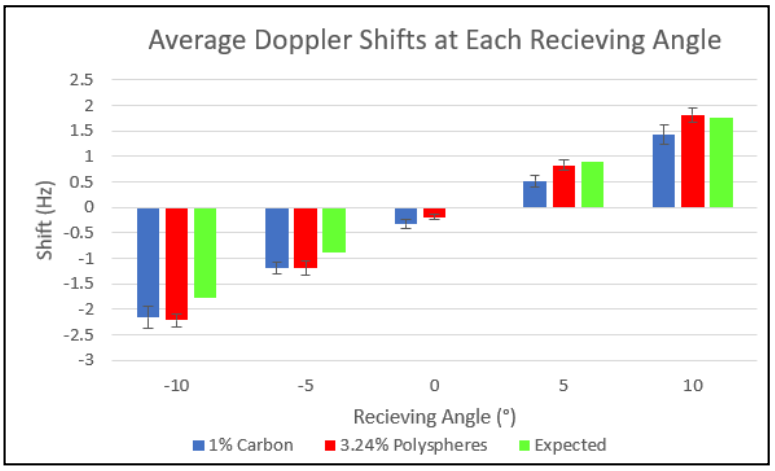

Figure 6.0: *Average doppler shift across our laser wavelength range of 680 - 800nm at each receiving angle compared to expected shifts calculated using equation 2.*

Figure 6.0 shows that PF gives greater magnitude shifts compared to the CF except at a receiving angle of 0°. Results for doppler shift of both fluids are shown to be larger in magnitude when the receiving angle is less than 0° compared to results above 0°. Results below 0° are shown to be less precise compared to results above 0°.

Table 3.1: *Doppler shifts of 1% carbon and 3.24% polysphere fluids averaged across the laser wavelength range of 680-880nm at each transducer receiving angle compared to the expected doppler shifts calculated using equation 2.*

Receiving Angle (°)	Expected Shift (Hz)	1% Carbon Shift (Hz)	3.24% Polyspheres (Hz)
10	1.763	1.420 ± 0.1859	1.81 ± 0.1418
5	0.8848	0.520 ± 0.1180	0.827 ± 0.1056
0	6.22×10^{-16}	-0.3333 ± 0.08548	-0.187 ± 0.05888
-5	-0.8848	-1.187 ± 0.1198	-1.19 ± 0.1457
-10	-1.763	-2.157 ± 0.223	-2.22 ± 0.1256

Table 3.1 shows that shifts for PF seen at receiving angles above 0° are shown to be 25.6% more accurate compared to shifts below 0°. Due to the 5° result for CF being 58.8% accurate, the average for above 0° accuracy is 2.1% less than below 0°. The PF is shown to give the most accurate results across all angles.

6.2 Signal-to-Clutter Ratio (SCR)

The following table and figure shows the average SCR for both fluids at each wavelength:

Table 4.0: *Average SCR across all transducer receiving angles at each laser wavelength for 1% carbon and 3.24% polyspheres fluids.*

Photoacoustic Fluid	Wavelength (nm)	Average SCR (dB)
1% Carbon	680	-8.142 ± 1.799
	720	-6.237 ± 0.9265
	760	-7.286 ± 1.243
	800	-11.31 ± 1.506
	840	-10.52 ± 1.711
	880	-9.855 ± 2.282
3.24% Polyspheres	680	-6.089 ± 1.256
	720	-6.110 ± 0.9579
	760	-7.631 ± 0.3149
	800	-7.154 ± 0.6776
	840	-5.221 ± 0.6383
	880	-8.119 ± 0.7006

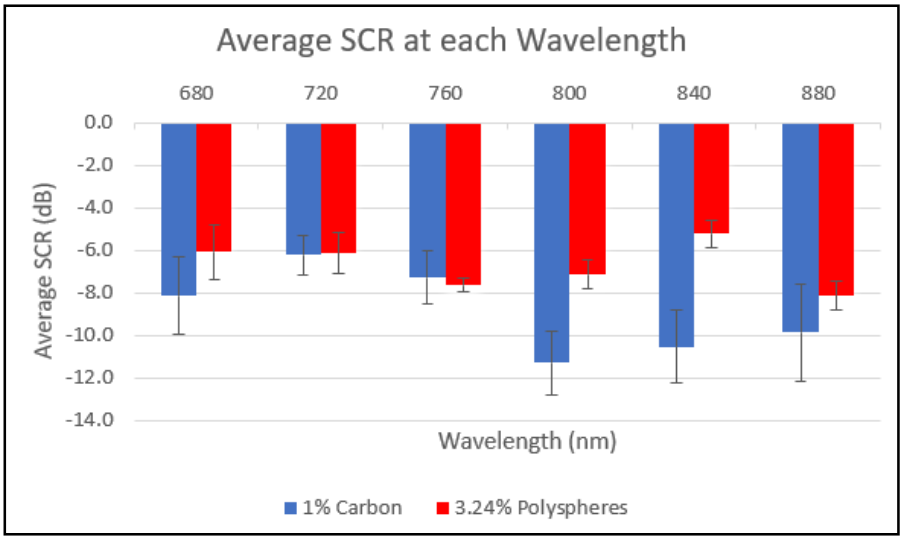

Figure 7.0: *Average SCR across all transducer receiving angles at each laser wavelength for 1% carbon and 3.24% polyspheres fluids plotted on a logarithmic scale.*

Table 4.0 and Figure 7.0 shows the PF has a less negative SCR compared to the CF at all wavelengths, with an exception at 760nm. Results produced by the PF are also shown to be more precise compared to results for the CF. The best result for SCR for the PF is -5.221 ± 0.6383dB at 840nm, and the CF is -6.721 ± 0.9265dB at 720nm.

Table 4.1: *Average SCR over our range of laser wavelengths for 1% carbon and 3.24% polyspheres.*

Photoacoustic fluid	SCR (dB)
1% Carbon	-8.892 ± 2.365
3.24% Polyspheres	-6.721 ± 1.256

These results reinforce findings in Table 4.0 and Figure 7.0, on average SCR is less negative and results are more precise for the PF.

6.3 Vector Flow

The following vector flow maps were constructed for both fluids at the associated wavelength that achieved the highest normalised intensity **(Table 3.0)**:

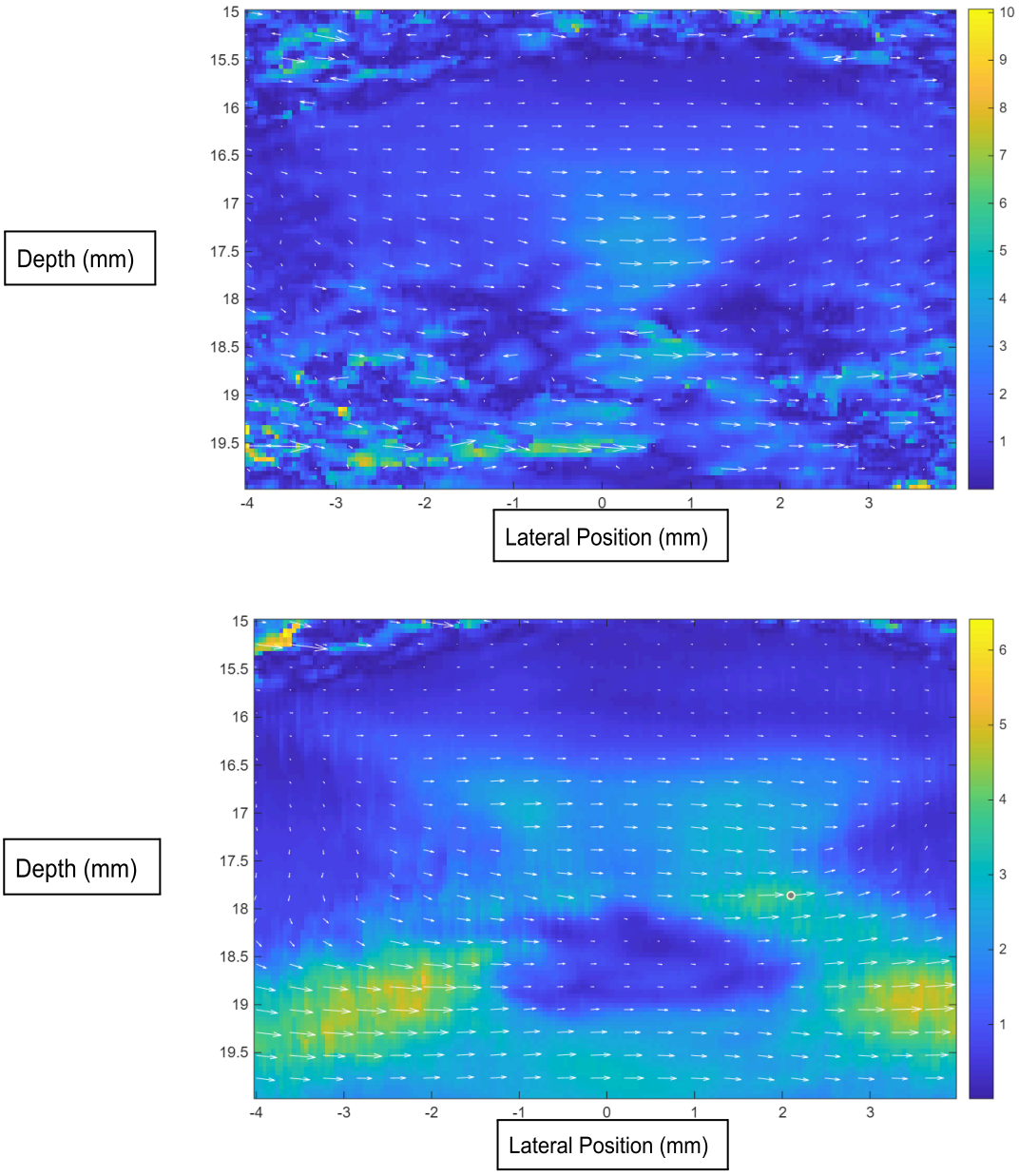

Figure 8.0: *Vector flow maps of 3.24% polysphere fluid (Top) and 1% carbon (Bottom) at laser wavelengths of 720nm and 880nm respectively.*

These vector flow maps show the direction and magnitude of flow through the phantom vein, with both maps showing laminar flow at depths between 16-18mm and lateral positions between -1-1mm. The flow map for the PF shows regions of inconsistent flow between depths of 15-15.5mm and 18-20mm. These regions are the edges of the vein/internal structure of the phantom where vector flow cannot be measured.

Table 5.0: *Mean flow velocities for both fluids over all transducer receiving angles at each laser wavelength.*

Fluid	Wavelength (nm)	Mean Flow Velocity (mm/s)
1% Carbon	680	1.9 ± 0.3414
	720	1.9 ± 0.3392
	760	2.1 ± 0.3301
	800	2.4 ± 0.3152
	840	1.9 ± 0.1985
	880	2.2 ± 0.2393
3.24% Polyspheres	680	2.5 ± 0.5096
	720	2.7 ± 0.5298
	760	2.7 ± 0.5799
	800	2.8 ± 0.6136
	840	2.6 ± 0.3792
	880	2.3 ± 0.4997

This figure shows our results from Table 5.0 compared against the expected mean flow velocity of 1.936 ± 0.0004mm/s:

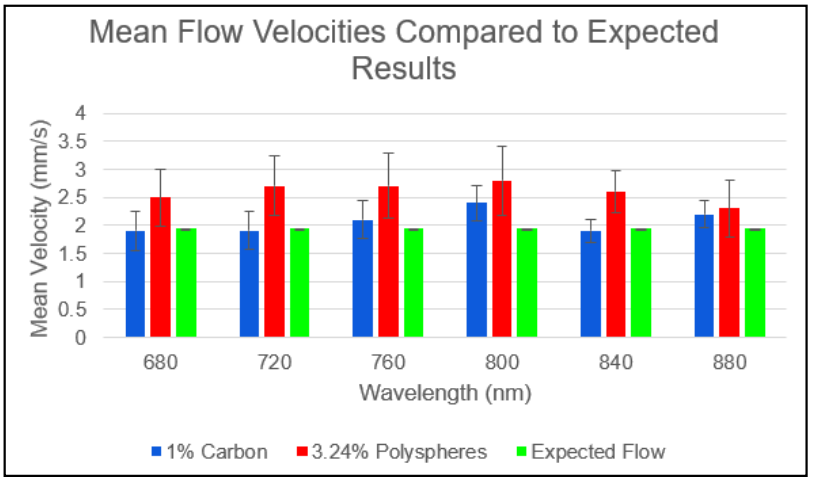

Figure 8.1: *Average velocity of flow seen by transducer at each wavelength ranging from 680-880nm for 1% carbon and 3.24% polysphere fluid compared to expected results.*

Figure 8.1 shows that the average flow velocity of PF is higher compared to both the CF and the expected results. PF is also shown to be less precise compared to CF. From Table 5.0 the most accurate flow velocity for the CF is 1.9 ± 0.1985mm/s at 840nm and for the PF is 2.3 ± 0.4997mm/s at 880nm.

Table 5.1: *Average velocity of flow for 1% carbon and 3.24% fluids through the phantom vein compared to the expected result.*

Photoacoustic fluid	Flow Velocity (mm/s)
Expected Flow	1.936 ± 0.004
1% Carbon	2.067 ± 0.2991
3.24% Polyspheres	2.600 ± 0.5238

Table 5.1 shows that both fluids over predict the expected flow, with the CF shown to be more accurate compared to the PF.

The Maximum flow velocities found at each wavelength are compared for both fluids in the following figure:

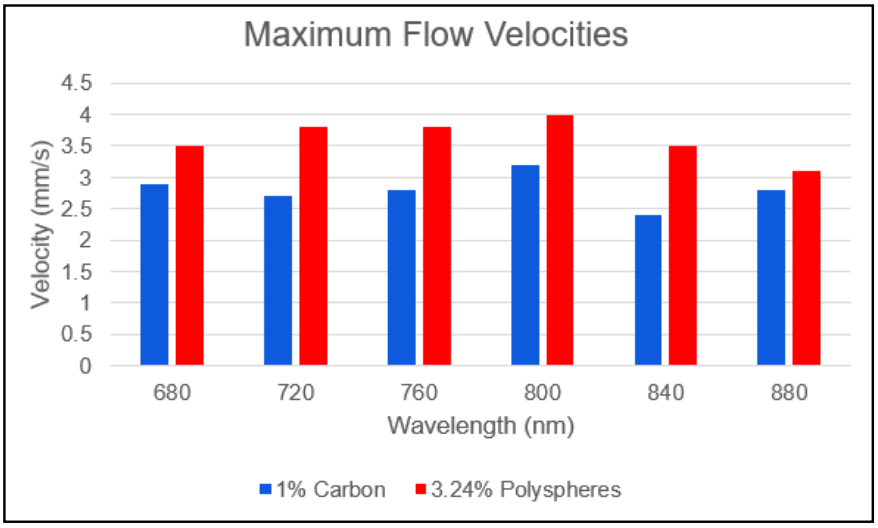

Figure 8.2: *Maximum flow velocities at each wavelength for 1% carbon and 3.24% polysphere fluid.*

The maximum flow velocity for PF is shown to be larger than the CF across all wavelengths.

7. Analysis of Results

7.1 Signal-to-Clutter Ratio Performance

7.1.1 Polysphere Fluid SCR

The PF demonstrated strong SCR performance across all wavelengths tested, with values ranging from -5.221 ± 0.6383 dB at 840nm to -8.119 ± 0.7006 dB at 880nm **(Table 4.0)**. The fluid achieved an overall average SCR of -6.721 ± 1.256 dB across the entire wavelength range (**Table 4.1)**, indicating consistently strong signal quality relative to background noise.

The PF showed relatively low variability in its SCR measurements, with a standard deviation of 1.256 dB. This consistency suggests stable optical properties and predictable performance across trials. The non-glassy surface of the polysphere particles likely contributes to this stability by scattering light more uniformly and reducing the intensity of noise-inducing reflections. This uniform scattering behaviour means that the photoacoustic signals generated are more consistent and less prone to hot-spots than particles with reflective surfaces.

The PF's strong SCR performance shows it would be well-suited for applications where image clarity and signal detection are critical, such as visualising fine vascular structures or distinguishing weak signals in challenging imaging conditions.

7.1.2 Carbon Fluid SCR

The CF demonstrated SCR values ranging from -6.237 ± 0.9265 dB at 720nm to -11.31 ± 1.506 dB at 800nm **(Table 4.0)**. The fluid achieved an overall average SCR of -8.892 ± 2.365 dB across the entire wavelength range (**Table 4.1)**

The CF showed considerable variability in its SCR measurements, with a standard deviation of 2.365 dB. This inconsistency across wavelengths suggests less stable optical behavior and more unpredictable performance. The glassy surface of the carbon particles likely contributes to this variability by creating reflections that vary depending on particle orientation and distribution within the vein. These reflections can introduce additional noise components that reduce the overall SCR and make measurements less consistent.

The black pigmentation of the carbon particles provides broad-spectrum absorption, which ensures signal generation across all wavelengths tested. The best SCR performance was observed at 720nm, where the signal amplitude was largest relative to the clutter peak. However, the overall lower and more variable SCR suggests challenges for applications requiring high image quality or detection of weak signals.

7.1.3 SCR Comparison and Implications

The PF demonstrated better SCR performance compared to the CF across all tested wavelengths. The PF achieved an average SCR of -6.721 ± 1.256dB **(Table 4.1),** with its best performance at 840nm reaching a SCR of -5.221 ± 0.6383dB **(Fig. 7.0).** The CF showed an average SCR of -8.892 ± 2.365dB **(Table 4.1),** with its best performance at 720nm reaching a SCR of -6.237 ± 0.9265dB **(Fig. 7.0)**.

This roughly 2dB difference in SCR is significant in PA. A higher (less negative) SCR means a stronger signal relative to background noise, which results in clearer images and more reliable measurements. The PFs' better SCR performance suggests it would produce higher quality images and more easily distinguishable flow signals in practical PAI applications.

The CF also showed increased variability in its SCR measurements, with a standard deviation of 2.365dB compared to 1.256dB for the polyspheres. This indicates less consistent performance across different wavelengths. This contradicts our initial hypothesis that carbon particles' stronger optical absorption would result in higher SCR values. The PF's more uniform optical properties and reduced light scattering from its non-glassy surface likely contribute to its more stable and superior SCR performance.

7.2 Flow Velocity Accuracy

7.2.1 Polysphere Fluid Flow Measurements

The PF showed flow velocity measurements ranging from 2.5 ± 0.5096mm/s at 680nm to 2.8 ± 0.6136mm/s at 800nm **(Table 5.0)**. The fluid achieved an overall average flow velocity of 2.6 ± 0.5238mm/s across all wavelengths **(Table 5.1)**, which represents a systematic overestimation of approximately 34% compared to the expected flow velocity of 1.936 ± 0.004mm/s. The most accurate measurement was 2.3 ± 0.4997mm/s at 880nm, which still overestimated the expected velocity by approximately 19%.

The PF showed relatively large uncertainty in its flow measurements, with standard deviations of approximately 0.5mm/s across all wavelengths tested. This increased uncertainty suggests that while the fluid produces strong photoacoustic signals, the translation of those signals into accurate velocity measurements is less reliable. The smaller particle size of the polyspheres (3µm compared to 7.5-8.7µm for RBCs) may contribute to this behaviour. The reduced particle size could affect doppler shift characteristics or create a fluid that can flow faster than expected by creating less friction than that of heavier and larger particles in the CF or blood [63,64].

The systematic overestimation of flow velocity across all wavelengths indicates a consistent bias in the measurements rather than random error. This could be related to the optical scattering properties of the polysphere particles affecting how the doppler shift is calculated, or potentially due to differences in how the red pigmentation interacts with the incident laser light. The reasons for this overestimation remain unclear and require further investigation.

7.2.2 Carbon Fluid Flow Measurements

The CF showed flow velocity measurements ranging from 1.3 ± 0.3096mm/s at 680nm to 2.8 ± 0.6136mm/s at 800nm **(Table 5.0).** The fluid achieved an overall average flow velocity of 2.067 ± 0.2991 mm/s across all wavelengths **(Table 5.1)**, representing a relatively small overestimation of approximately 6.8% compared to the expected flow velocity of 1.936 ± 0.004mm/s. The most accurate measurement was 1.9 ± 0.1985mm/s at 840nm, which was within 1.86% of the expected value

The CF showed reasonable uncertainty in its flow measurements, with standard deviations of roughly 0.3mm/s across most wavelengths tested. This level of uncertainty suggests relatively consistent velocity measurements across the wavelength range. The larger particle size range of the carbon particles (2-12μm) may contribute to this behaviour, as particles closer to RBC dimensions (7.5-8.7μm) could produce doppler shifts more representative of actual blood flow.

Figure 7.1 shows a strong signal of flow for the CF at depths between 18 - 20mm. This is likely due to the reflections in the CF, due to its glassy texture, bouncing between particles before heading to the US transducer. Therefore taking longer to arrive and thus being positioned lower in the figure.

The CF showed relatively small deviation from the expected flow velocity across most wavelengths. The close agreement at 840nm suggests that the carbon particles' optical and acoustic properties translate well into accurate flow velocity quantification through the doppler shift calculations. The black pigmentation and broad-spectrum absorption of the carbon particles appear to produce photoacoustic signals that accurately represent velocity information despite variations in signal strength across wavelengths.

7.2.3 Flow Accuracy Comparison and Implications

When comparing the two fluids, the CF demonstrated significantly better accuracy than the PF across the entire wavelength range tested. As noted in the previous sections, the CFs overall deviation of 6.8% compared favourably to the polyspheres 34% deviation, and the CFs best performance at 840nm (within 1.86%) was significantly better than the polyspheres best result at 880nm (19% overestimation).

The PF also exhibited considerably larger uncertainty in its measurements, with the standard deviations being nearly double that of the CF. This suggests that while the PF produces clearer signals (as shown in the SCR results), these signals may be less accurate for quantitative velocity measurements. This presents an interesting trade-off between image quality and measurement accuracy. Similar observations have been reported by Brunker and Beard (2016), who found that polystyrene microsphere suspensions consistently underestimated flow velocities across a range of vessel widths when using photoacoustic Doppler velocimetry [45]. The reasons for the PF's systematic overestimation of flow velocity are unclear but could potentially be related to its smaller particle size or its optical scattering properties affecting the doppler shift calculations.

7.3 Wavelength-Dependent Performance

7.3.1 Polysphere Fluid Wavelength Response

The PF demonstrated wavelength-dependent performance characteristics that reflect its design to mimic the optical properties of blood. PFs have optical properties that better represent RBCs in vivo due to absorption spectra of polystyrene being similar to that of oxygenated RBC haemoglobin and the red dye used to colour the spheres being similar to that of the pigment of RBCs. The absorption coefficients of RBCs and a 3µm polyspheres solution is shown in the supplementary material J. Brunker and P. Beard (2016) [45]:

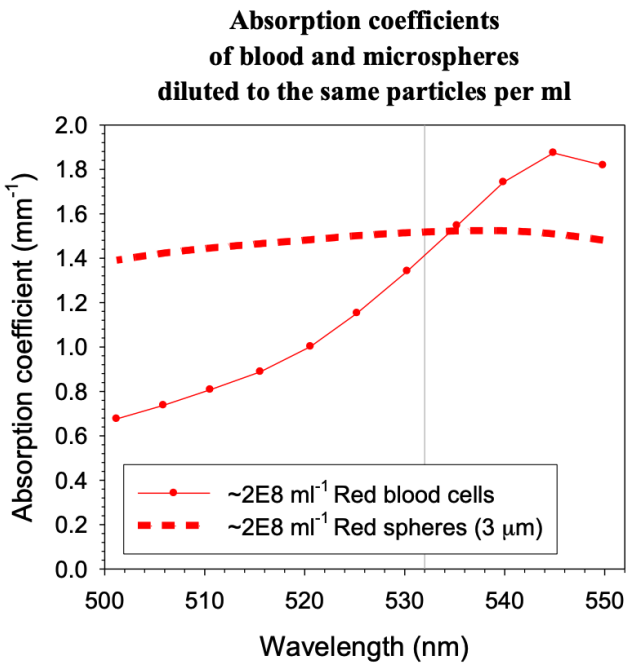

Figure 9.0: *Absorption coefficients for 3µm polyspheres solution compared to RBCs with equivalent concentrations in de-ionized water over a wavelength range of 500-550nm.*

In Figure 9.0 it shows the absorption coefficients agree between 530-540nm suggesting that in this particular range the absorption properties of the two fluids are very similar. This study did not test this wavelength, as it does not fall in the range 680-880nm.

While the experimental wavelength range did not overlap with the optimal matching region shown in Figure 9.0, it shows that polysphere mimics blood absorption into the near-infrared range, which was tested. Blood haemoglobin shows increased absorption in the near-infrared range above 700nm, which provides context for the PF's performance across our 680-880nm testing range [65]. The fact that polyspheres are designed to replicate blood's spectroscopic behaviour means their wavelength-dependent performance should reflect physiologically relevant absorption characteristics, even at wavelengths where direct comparison data is unavailable.

When examining the normalized signal intensity data **(Table 3.0)**, the PF showed values ranging from 0.5512 at 880nm to 1.866 at 720nm with a standard deviation of 0.43. This consistency across the

near-infrared spectrum reflects the stability of the red pigment's optical properties and suggests that the fluid maintains predictable absorption behavior throughout the wavelength range relevant to photoacoustic blood flow imaging.

7.3.2 Carbon Fluid Wavelength Response

The carbon particles' black pigmentation provides broad-spectrum absorption across the visible and near-infrared range. This non-selective absorption means the carbon particles generate photoacoustic signals across all tested wavelengths, from 680nm to 880nm.

When examining the normalized signal intensity data **(Table 3.0)**, the CF showed a general trend of increasing normalized intensity at longer wavelengths, ranging from 16.60 at 760nm to 60.67 at 880nm with a standard deviation of 16.4. The particularly high normalized intensities at 840nm (55.89) and 880nm (60.67) suggest that the carbon particles' absorption efficiency improves at longer wavelengths, or that the reduced scattering at longer wavelengths allows more effective signal generation [66].

However, the CF exhibited considerable variability in its normalized intensities across the wavelength range. The drop to 16.60 at 760nm, sandwiched between higher values at 720nm (29.15) and 800nm (31.98), indicates inconsistent wavelength-dependent behavior. This variability likely stems from the glassy, reflective surface of the carbon particles creating wavelength-dependent scattering effects. These reflections introduce noise components that vary with wavelength, reducing the predictability of the CF's performance across the spectrum.

The variable particle size range of the carbon particles (2-12μm) may also contribute to this wavelength-dependent behavior. Different sized particles within this range interact differently with photons of varying wavelengths, potentially creating complex interference patterns that affect both signal generation and acoustic wave propagation [67]. This size heterogeneity introduces additional complexity into how the CF responds across different wavelengths, making its performance less predictable than a fluid with uniform particle sizing.

7.3.3 Optimal Wavelength Selection

Both fluids showed distinct wavelength-dependent performance characteristics, with different optimal wavelength for SCR and flow accuracy measurements. For the PF, 840nm provided the best SCR performance, while 880nm yielded the most accurate flow measurements. The CF showed its best SCR at 720nm and most accurate flow velocity at 840nm.

The difference in optimal wavelengths between the two fluids likely stems from their distinct optical properties. The PF's red pigmentation is designed to mimic blood's absorption spectrum. Blood haemoglobin shows strong absorption peaks in the visible range at 540 and 576nm for oxygenated haemoglobin but also exhibits increased absorption in the near-infrared range above 700nm [65]. This aligns well with the

polysphere's best performance at 840-880nm. This study didn't observe performance at lower wavelengths because the testing range started at 680nm. Testing at shorter wavelengths corresponding to blood's primary absorption peaks could potentially yield different optimal wavelengths. The CF's black pigmentation provides broader absorption across the spectrum but showed optimal performance at the shorter 720nm wavelength for SCR, suggesting different absorption characteristics compared to blood.

Interestingly, 840nm emerged as a particularly important wavelength in our experiments. It provided the best SCR for polyspheres and the most accurate flow measurements for carbon. This suggests that 840nm could be a good compromise wavelength when switching between fluids or prioritising different measurement goals. However, the fact that optimal wavelengths differ between SCR and flow accuracy measurements for the same fluid highlights the complexity of selecting experimental parameters in PAI. Depending on whether image quality or measurement accuracy is more critical for a given application, different wavelengths should be selected accordingly.

7.4 Overall Comparative Analysis

When evaluating both fluids across all metrics, it becomes clear that neither fluid emerges as a definitive winner for all photoacoustic imaging applications. Instead, each fluid presents distinct trade-offs in performance characteristics.

The PF has advantages for signal quality, producing clearer images with better SCR performance across all wavelengths tested (average of -6.72dB compared to -8.89dB for CF). This superior signal strength translates to better image clarity and more easily distinguished flow signals. However, this advantage in signal quality comes at the cost of flow measurement accuracy, with velocities consistently overestimated by ~34% compared to expected values.

The CF demonstrates the opposite performance profile. It achieves superior accuracy for flow velocity measurements, with readings within 6.8% of expected values overall and near-perfect accuracy (within 1.86%) at 840nm. This makes it significantly more reliable for applications requiring precise velocity quantification. However, it produces weaker and less consistent signals, with both lower average SCR and greater variability across wavelengths compared to polyspheres.

The fundamental trade-off between these fluids is clear: signal quality vs flow measurement accuracy. The PF provides better images but less accurate data, while the CF provides more accurate measurements but with weaker signals.

8. Error Analysis/Limitations

8.1 Systematic Errors

8.1.1 Wavelength-Dependent Laser Power Variations

A significant source of error in our photoacoustic measurements arose from wavelength-dependent variations in laser output power. Tunable laser systems tend to exhibit non-uniform power profiles across their operation wavelength range due to gain medium efficiency and properties [68,69]. These power fluctuations directly affect the amplitude of photoacoustic signals, potentially causing non-normality amongst our results. This was an important source of error throughout the experiment. To ensure this was taken into account, power measurements were taken for each of the wavelengths before doing the experiment, allowing correlation with the results and ensuring no abnormalities arose from this specific issue.

8.1.2 Laser Alignment Uncertainty

The absence of an analytical method for laser alignment introduced potential limitations in experimental reproducibility. In PAI, precise spatial overlap between the optical excitation and acoustic detection regions is essential to ensure consistent signal amplitude and image quality. In this study, alignment was performed empirically by iteratively adjusting the laser position to maximise the measured signal amplitude and then fixing it in place at the observed maximum. While sufficient for single-session measurements, this approach reduces repeatability across multiple experimental sessions. Minor misalignments accumulated over time could significantly affect data reliability and comparability. Although this limitation did not impact our results, as all trials were conducted within a single day, the adoption of an analytical alignment procedure is recommended for experiments requiring repeated measurements.

8.2 Sample Stability and Preparation

8.2.1 Carbon Particle Settling

Carbon particle settling was both an immediate experimental concern and a long term stability issue that could affect measurement reproducibility and slow down experimentation. The carbon particles in the CF have a higher density than the carrier fluid, causing them to settle over time due to gravity. This causes a gradient of carbon and clear carrier fluid within the phantom and experimental setup. This introduces potential measurement issues; specifically, if the carbon particles fall out of suspension, the PAI system records the carrier fluid rather than the intended medium. To minimise this source of error, the carbon suspension was thoroughly mixed between measurements and monitored to ensure particle suspension during data acquisition. This is a relatively small issue, however it does raise some concerns with experimental viability, if time is taken during the experiment to re-mix the fluid between each measurement, using a blood

mimicking fluid may be less convenient than thought. This is a major practical limitation in using the CF for blood mimicking and should be taken into account. In future experiments, a stabilising agent such as xanthan gum could be used to prevent sedimentation and improve the practicality of the fluid. Xanthan gum has been successfully used in blood-mimicking fluids to maintain particle suspension while providing rheological properties similar to blood [70]. However, it would be necessary to characterise how such additives affect the optical absorption and acoustic properties of the suspension, as changes to the fluid's viscosity or acoustic impedance could impact the accuracy of photoacoustic measurements.

8.2.2 Microsphere Staining

The polystyrene microsphere fluid used was red in colour to better mimic the optical properties of blood. After conducting trials with the poly-sphere fluid,staining was found on the outer edges of our phantom vessel. This presents some major issues with running multiple experiments/trials and repeatability. If the outer edges of the phantom become too stained they can effectively “steal” the absorption from the flowing liquid and can result in measurement of the stagnant staining on the outer edges, making the results less reliable and inaccurate. This is because the dye molecules from the fluid bind to the phantom walls over time. Once bound, these molecules become stationary absorbers that contribute to the photoacoustic signal alongside the flowing fluid. Since the signal from the stained walls can't be separated from the flowing fluid, our measurements of the fluid's properties may become less accurate. This was not a significant issue for this study's limited number of trials, but could become problematic with extended use. For future repeatability, researchers may have to switch between phantoms or develop a way of removing the staining from the inner vein between experiments.

8.2.3 Practical Limitations and Cost Considerations

Beyond performance metrics, the practical usability of each fluid presents significant limitations that impact their suitability for different experimental contexts. The PF, while offering superior signal quality, comes at a substantial cost of $635 per 5mL, roughly 7 times more expensive than the CF [17]. This cost difference can be prohibitive for laboratories conducting numerous trials or working with limited budgets, potentially restricting access to what is otherwise the better performing fluid in terms of image quality.

The carbon particle settling issue, as discussed in section 8.2.1, presents a major practical limitation during experimentation. The constant need to remix the fluid between measurements slows down the experimental process and introduces potential for inconsistency if mixing is insufficient. The handling requirement makes the CF less convenient for extended measurement sessions or high volume experiments, despite its superior accuracy for flow measurements.

Similarly, the microspheres staining problem discussed in section 8.2.2 limits the long term repeatability of experiments using the PF. As the phantom walls become stained over time, they act as stationary absorbers that cannot be distinguished from the flowing fluid signal, progressively degrading

measurement accuracy. This necessitates either frequent phantom replacement (adding additional cost) or developing cleaning procedures, both of which add complexity to experimental workflows.

These practical limitations highlight that the choice between fluids cannot be made on performance data alone. The experimental context, including budget constraints, number of trials required, and timeframe for measurements, must be considered alongside the optical and flow measurement performance when selecting an appropriate blood mimicking fluid.

8.3 Signal Processing and Measurement Artifacts

8.3.1 Aliasing in Sampling

Aliasing was an issue to consider with this experiment. When measuring vector flow it is important that the rate of flow is slow enough to not get any aliasing and noise artifacts in the measured signal. If the flow velocity is too high relative to our measurement rate, the velocity information can fold back and appear as flow in the wrong direction or at incorrect speeds [71]. To avoid this, the flow rate was kept within a range where aliasing wouldn't occur by setting our flow rate to be less than half of the maximum measurable velocity (Vmax) for our given measurement system. Increasing the pulse rate of the laser increases the nyquist frequency thus enabling faster flows to be recorded.

8.3.2 Environmental Noise

Photoacoustic measurements are sensitive to acoustic interference from environmental sources. This can include vibrations from building infrastructure, air conditioning systems, and electromagnetic interference. These noise sources add unwanted components to our signal that reduce the SCR, particularly for weak signals from low absorber concentrations or at higher imaging depths. To mitigate these effects, the experimental setup incorporated several noise reduction measures. An isolation table was used to decouple the experiment bench from seismic and building vibrations. Additionally, a black cloth was placed over the setup to limit acoustic reflections from the metal base under the phantom, reducing secondary acoustic interference.

Despite these precautions, residual noise components from stationary sources (visible as the clutter peak around 0Hz in the frequency spectrum) remained in the photoacoustic signals. During data processing, a high-pass filter with a cut-off frequency of 0.01Hz was applied to remove this clutter peak, effectively isolating the Doppler signal from the moving absorbers. This filtering allowed accurate measurement of flow velocities by distinguishing the signal produced by flowing particles from the noise generated by stationary components of the experimental setup. In future experiments, active noise cancellation systems could further improve real-time measurement accuracy by reducing the need for post-processing filtering.

9. Future Considerations

The results of this study call for future research to be conducted in this field. It has been shown that the raw data from the US transducer does not completely reflect reality and calibration coefficients must be taken into consideration [2]. This coefficient will alter the data presented in this study which should be considered by future researchers. In addition, it is shown in Figure 6.0 that receiving angles under 0° experience larger doppler shifts than their positive counterparts. If this is consistent in future research, we recommend creating a weighting calibration that corrects for this so that the Doppler shifts are even, hopefully leading to more accurate velocity calculations across all angles.

10. Conclusion

This study compared two blood-mimicking fluids for photoacoustic vector-flow imaging: a 1% carbon sphere fluid (CF) and a 3.24% polystyrene microsphere fluid (PF). Testing was conducted across wavelengths from 680 to 880nm using a 2.95mm diameter phantom vessel. The results revealed a fundamental trade-off between signal quality and measurement accuracy. The PF demonstrated superior SCR ratio performance, achieving an average SCR of -6.7 ± 1.3dB compared to the CF's -8.9 ± 2.4dB. This translates to clearer images and more easily distinguishable flow signals, making it well suited for applications prioritising image quality. The CF provided significantly better flow velocity accuracy, with measurements within 7% of expected values compared to the polyspheres 34% deviation. At 840nm, the CF achieved near perfect accuracy (within 1.86%), demonstrating its reliability for quantitative flow measurements. Beyond performance metrics, significant practical limitations were identified. The CF requires constant remixing due to particle settling, while the PF stains phantom walls and is substantially more expensive. These factors critically impact experimental workflow, reproducibility, and cost-effectiveness.

In summary, the choice between these BMFs is application-dependent. For qualitative imaging and visualization of vascular structures, the 3.24% polystyrene fluid is recommended due to its superior SCR. For quantitative studies requiring precise flow velocity measurements, the 1% carbon fluid is the more reliable option, despite its lower signal strength and handling inconveniences. This work underscores the critical need for continued development of standardized, multi-parameter BMFs that balance optical absorption, acoustic properties, and practical usability to accelerate the clinical translation of photoacoustic imaging technologies.

11. Authors Contribution

M.E, L.K and G.P designed the experimental methodology, fabricated the phantoms and carried out the experiment including image processing. L.K. performed the data processing and analysis using MATLAB and Python. M.E wrote the Introduction, Literature Review, drew the diagrams of experimental equipment, and coordinated communication with the laboratory supervisor. L.K wrote the Data Processing and Results Section. G.P. wrote the Method, Discussion, Limitations and Conclusion sections.

All authors contributed equally to the experimental methods, reviewed and edited the manuscript, and approved the final version. All authors were involved in all sections of the report, data collection, processing and experimentation, the above description is the "main" contributions for each person.

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